Novara Hospital

Key Business Objectives

1.Encounters Overview

Cost and Coverage Insights

Patient Behaviour Analysis

Business Problem

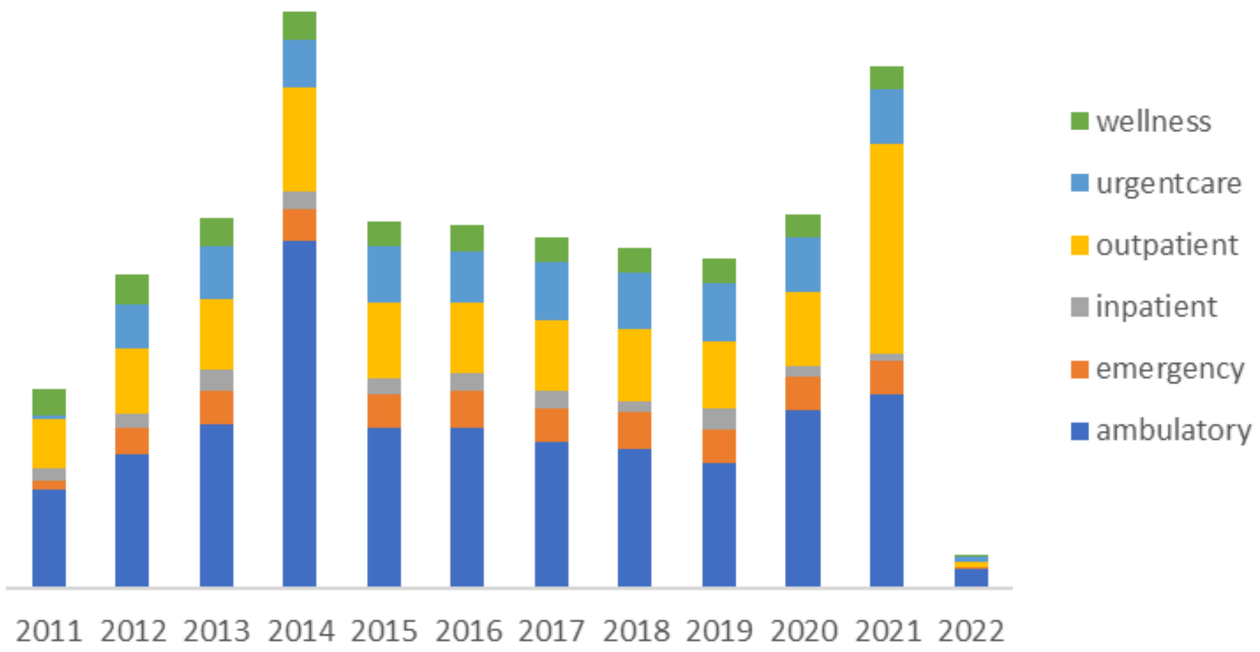
- Hospitals generate large volumes of patient encounter data, but without structured analysis, it is difficult to identify patterns in patient visits, cost drivers, insurance coverage gaps, and readmission trends.
- As a result, decision-makers may struggle to optimize resource allocation, control costs, and improve patient outcomes.
- This project aims to bridge that gap by transforming raw hospital data into actionable insights that support operational efficiency, financial planning, and quality of care.

Encounter Class Distribution Over time

- Ambulatory & outpatient dominate (>40%)
- Stable trend from 2011–2020
- Shift in 2021: outpatient > ambulatory
- Ambulatory : ~42%
- Outpatient: ~20%

Result Grid | Filter Rows:

cnt	encounterclass
12529	ambulatory
6287	outpatient
3664	urgentcare
2321	emergency
1920	wellness
1135	inpatient



Short vs Long duration encounters

- Under 24 hours : ~96%
- Over 24 hours: ~4%

Result Grid | Filter Rows: | Expo

stay_category	total_encounters	pct_enct
Under 24 hrs	26704	95.8644
OVER 24 hrs	1152	4.1356

Insights:

- Ambulatory encounters account for ~42%, while outpatient encounters contribute ~20% of total encounters, making them the dominant encounter classes.
- Additionally, ~96% of encounters are completed within 24 hours, indicating a high volume of short-duration cases.

Interpretation

- These patterns suggest that the hospital primarily focuses on short-term, non-critical care services, with limited involvement in long-duration or high-acuity treatments.

Recommendations:

- The hospital should prioritize resource allocation toward ambulatory and outpatient services, including staffing, scheduling, and infrastructure optimization.
- Given the high proportion of short-duration encounters, improving patient throughput and operational efficiency can further enhance service delivery.

Proportion of Encounters with Zero payer coverage

- Approximately 49% of encounters have zero payer coverage, indicating that nearly half of all patient visits are not supported by insurance.

Interpretation

This suggests a substantial reliance on out-of-pocket payments or potentially unrecovered costs, exposing both patients and the hospital to financial risk. Such a high proportion of uncovered encounters may impact revenue stability and accessibility of care.
This indicates a potential gap in healthcare accessibility and insurance penetration among patients

Result Grid | Filter Rows:

total	total_pct
13577	48.7399

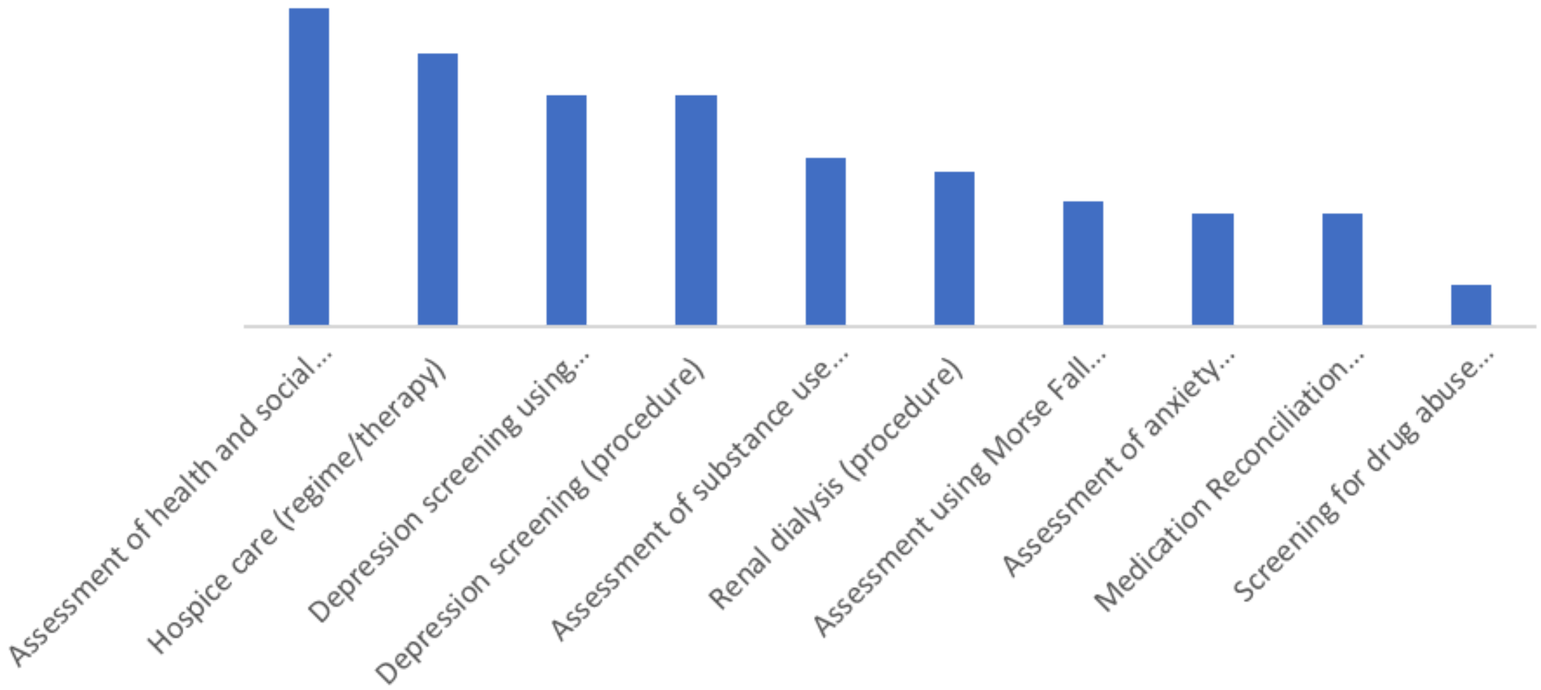
Recommendations

The hospital should:

- Strengthen insurance partnerships and network coverage
- Improve insurance verification at admission
- Identify patient segments with low coverage and design financial assistance or payment plans
- Monitor uncovered encounters to reduce potential revenue leakage

Top 10 Procedures by Volume with Average Cost

- **Preventive and assessment procedures dominate in volume**
- **“Assessment of health and social care needs” is the most frequently performed procedure**
- **High-volume procedures tend to have relatively lower average costs, indicating operational efficiency in routine services.**



Interpretation:

- High frequency of screening and evaluation procedures indicates a focus on early diagnosis and routine care
- These procedures are generally low to moderate in cost

Recommendations:

- Optimize workflows for high-volume procedures to improve efficiency
- Standardize processes to reduce operational cost
- Invest in preventive care programs to maintain early detection advantage

Most Expensive Procedures and their Frequency

- Admit to ICU has the highest average base cost, but is performed very infrequently (5 times).
- In contrast, **Electrical cardioversion** is performed frequently (~1383 times) while still ranking among the top high-cost procedures.

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

description	cnt	avg_b_cost
Admit to ICU (procedure)	5	206260.4000
Coronary artery bypass grafting	9	47085.8889
Lumpectomy of breast (procedure)	5	29353.0000
Hemodialysis (procedure)	27	29299.5556
Insertion of biventricular implantable cardioverter defibrillator	4	27201.0000
Electrical cardioversion	1383	25903.1106
Partial resection of colon	7	25229.2857
Fine needle aspiration biopsy of lung (procedure)	1	23141.0000
Percutaneous mechanical thrombectomy of portal vein using fluoroscopic guidance	57	20228.0351
Percutaneous coronary intervention	9	19728.0000

Interpretation:

- High-cost procedures like ICU admissions are rare but resource-intensive, contributing significantly to per-case cost.
- Procedures like electrical cardioversion represent a unique category of high-cost yet high-volume services, indicating consistent demand despite relatively higher costs.
- This indicates that not all high-cost procedures are rare—some contribute significantly to both cost and operational workload.

Recommendations:

- Closely monitor high-cost, low-frequency procedures (like ICU admissions) for cost control and resource optimization.
- For high-volume, high-cost procedures (like electrical cardioversion):
- Improve operational efficiency
- Optimize scheduling and resource allocation
- Explore cost standardization opportunities

Final Insights :

- **Preventive care dominates volume(Assessment care)**
- **Critical care dominates cost(Admit to ICU)**

Average Claim Cost by Payer

- Medicaid has the highest average claim cost (~6205), followed by uninsured patients (~5597)
- Private insurers (Anthem, Humana, etc.) show moderate average costs
- Medicare and Dual Eligible segments have comparatively lower average claim costs

Result Grid | Filter Rows:

p_name	avg_claim_cost
Medicaid	6204.885707
NO_INSURANCE	5596.793639
Anthem	4236.811151
Humana	3269.300148
Blue Cross Blue Shield	3245.584876
Cigna Health	2999.136431
UnitedHealthcare	2848.342200
Aetna	2758.769979
Medicare	2168.360510
Dual Eligible	1696.190779

Interpretation:

- Higher costs for Medicaid may indicate:
  - coverage of high-risk or complex patient groups
  - higher dependency on healthcare services
- High costs for uninsured patients suggest:
  - delayed treatment, leading to more severe conditions
  - potential financial burden or unrecovered revenue
- Lower costs for Medicare and private insurers may indicate:
  - better cost regulation
  - more structured treatment pathways

Recommendations:

- Focus on cost management strategies for:
  - Medicaid patients
  - uninsured segment
- Strengthen:
  - early intervention programs
  - insurance coverage awareness
- Analyze payer contracts to:
  - improve cost efficiency
  - optimize reimbursement structures

Quarterly Trend of unique patients admissions

- Patient admissions show a generally stable trend across quarters
- Slight variations observed between quarters within each year
- No strong or consistent seasonal spike across years
- Peak admissions were observed in Q1 of 2014 and 2021

Result Grid | Filter Rows:

yr	qut	no_of_pat
2011	1	156
2011	2	161
2011	3	155
2011	4	168
2012	1	249
2012	2	260
2012	3	232
2012	4	240
2013	1	242
2013	2	270
2013	3	259
2013	4	242
2014	1	394
2014	2	314
2014	3	272
2014	4	260
2015	1	244
2015	2	258
2015	3	251

Interpretation:

The hospital experiences relatively consistent patient inflow throughout the year, indicating stable demand rather than strong seasonal dependency.

Recommendations:

Maintain consistent resource allocation across quarters while monitoring minor fluctuations to optimize staffing and operational efficiency.

30-Day Patient Readmission Analysis

- Total 30-day readmissions: 17,308
- Repeated readmissions concentrated among a few patients

Result Grid | Filter Rows:

	readmitted_patients_count
▶	17308

Patients with most readmissions:

- **Repeated readmissions concentrated among a few patients**
- **Indicates pattern beyond random occurrences**

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

patient	readmitted_patients_count
1712d26d-822d-1e3a-2267-0a9dba31d7c8	1376
3de74169-7f67-9304-91d4-757e0f3a14d2	876
5e055638-0dad-dfd5-005d-1e74b6fd29ac	871
3f523789-55f3-bb31-2757-4803ca6a9c2a	442
5dcb295d-92df-a147-ebcc-aa49b6262830	421
442dc617-c7f2-0513-15cd-c35c4efdba73	384
b4ab9ab3-f52a-751e-a990-3bd5654d5870	376
9b8ae606-5059-b3a6-19c7-c812901898bb	347
b4671e80-7e87-9260-ca31-6a9397d465d1	289
ff331e5c-ab16-e218-f39a-63e11de1ed75	280

Recommendations:

- Identify and track high-risk patients (frequent readmissions)
- Strengthen post-discharge follow-up programs
- Introduce preventive care plans for chronic conditions

Impact:

- This analysis provides a clear view of hospital operations, patient behaviour, and financial patterns, enabling more informed decision-making.
- Improves visibility into high-volume services and resource utilization
- Highlights gaps in insurance coverage affecting revenue
- Identifies high-risk patients contributing to repeated readmissions
- Supports cost optimization for high-impact procedures

- **Outcome:**
- Enables the hospital to enhance operational efficiency, reduce avoidable readmissions, and improve both patient care quality and financial performance.